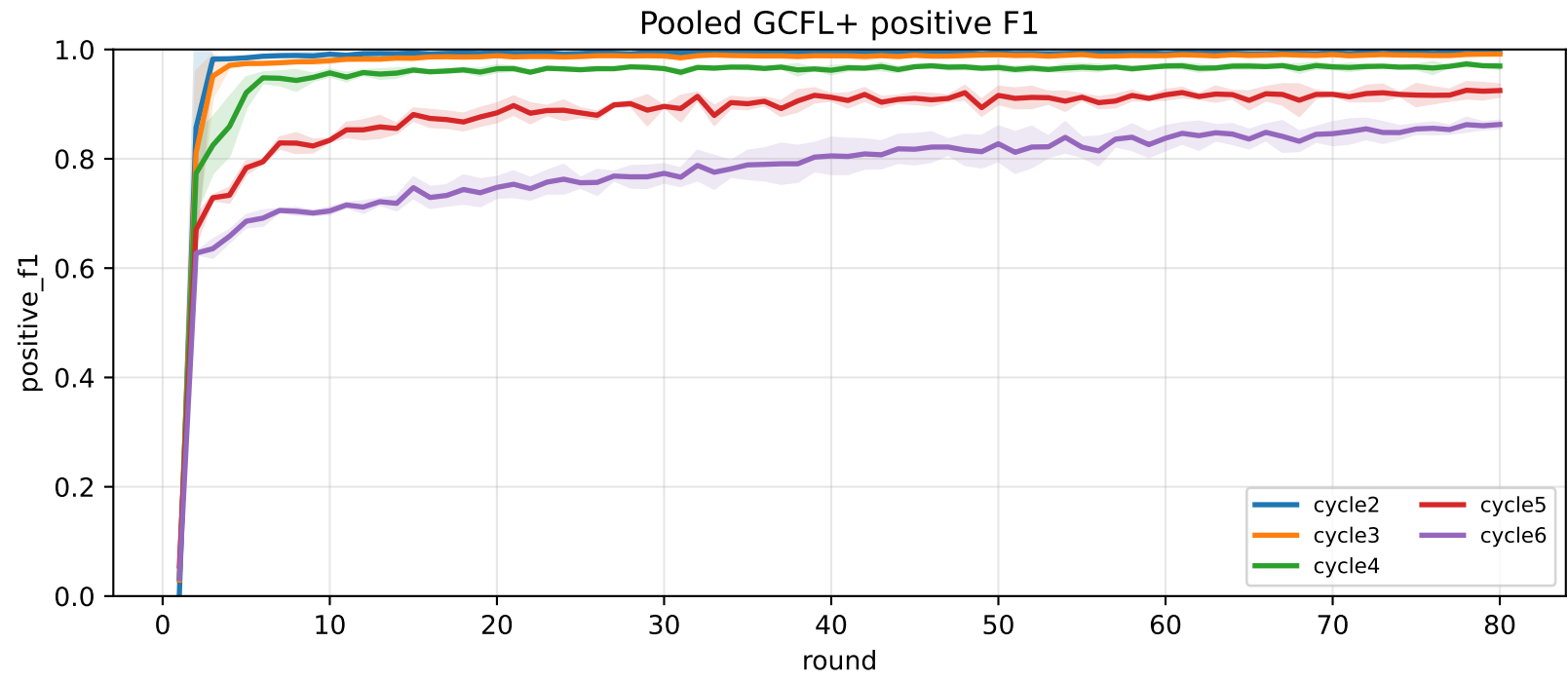
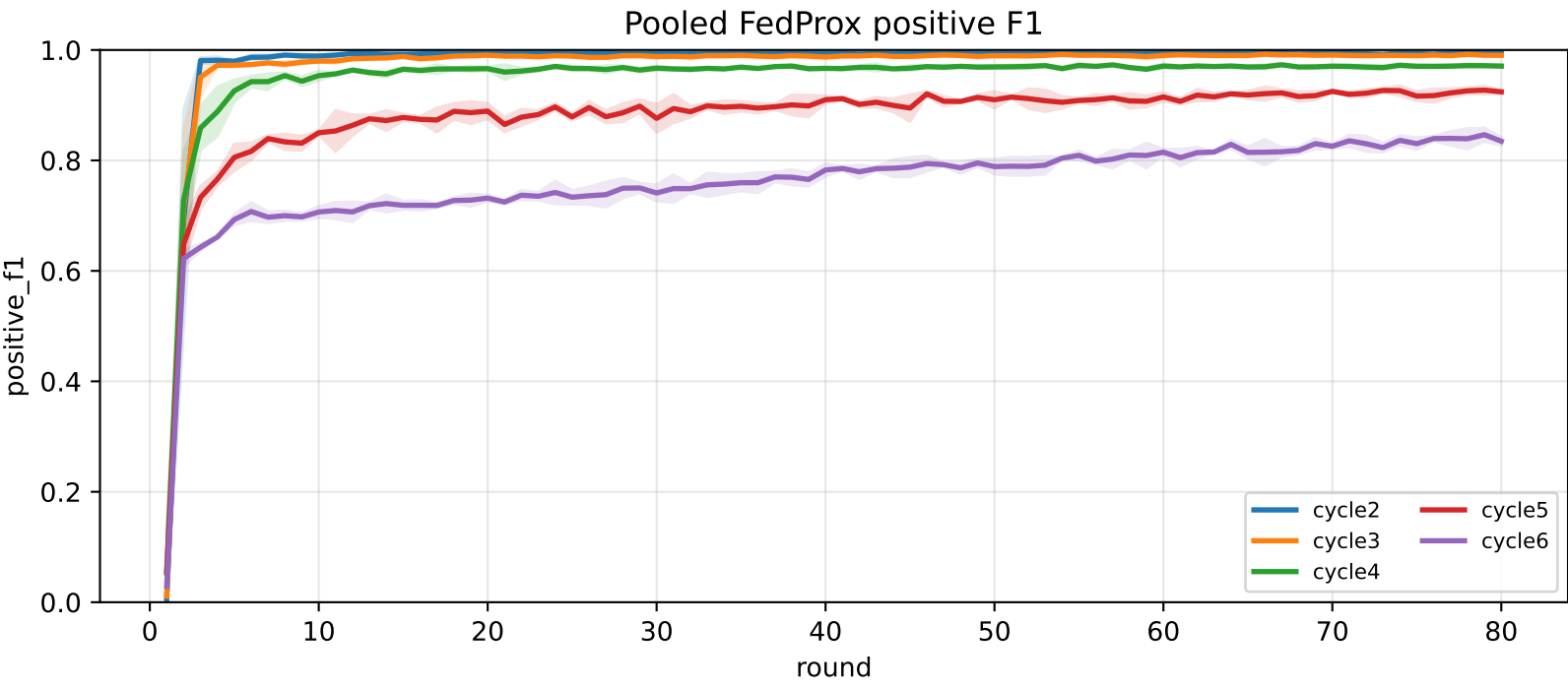
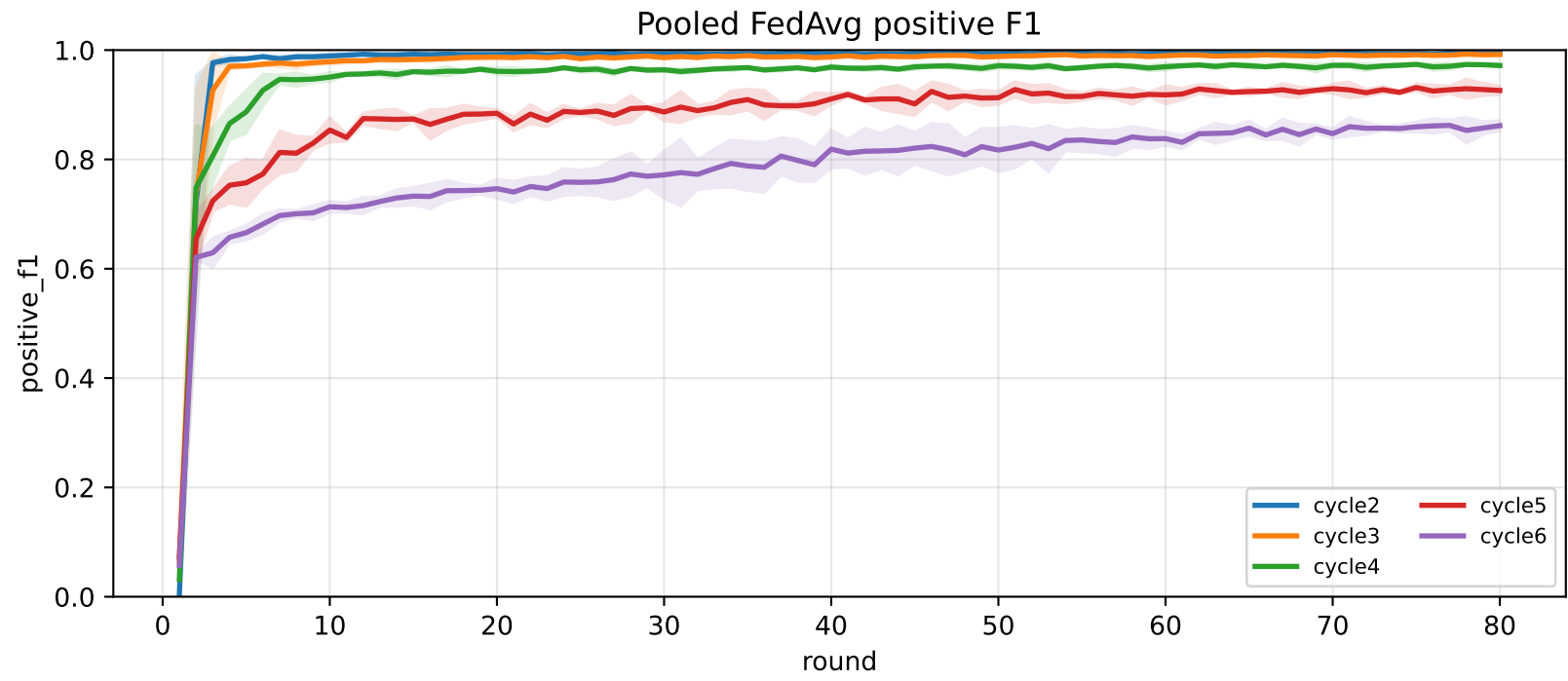
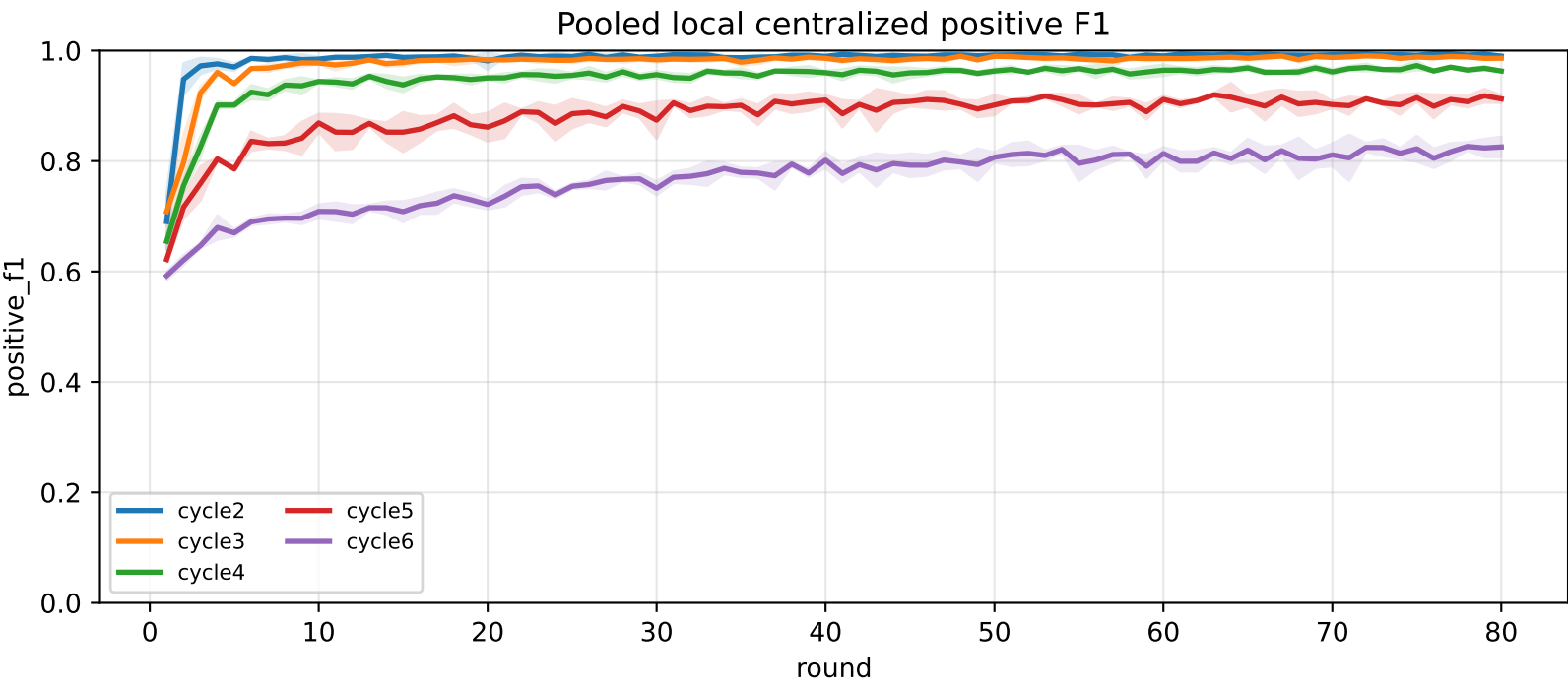
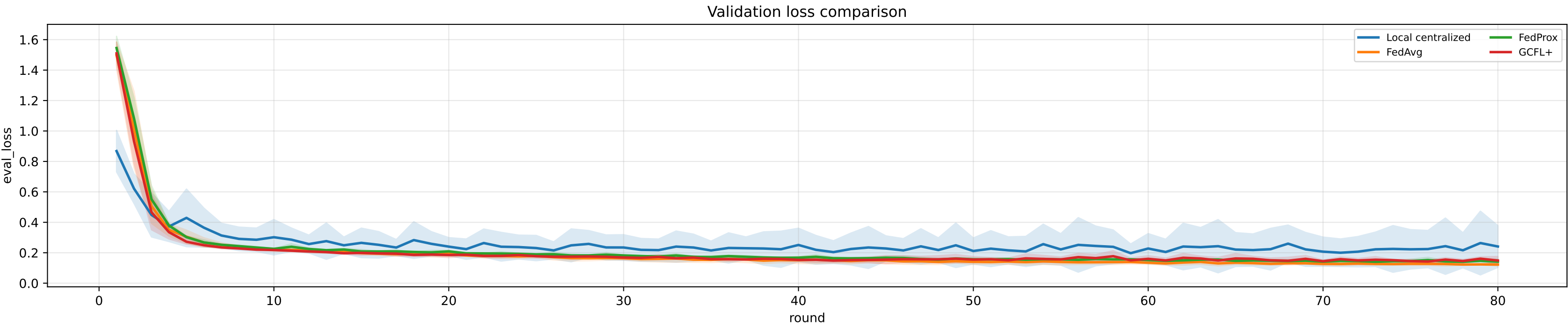


Controlled masking level: High | aggregated / pooled performance over 5 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 | family_centroid_jsd_mean=0.069

Best logged test performance (mean $\pm$ std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.6% $\pm$ 0.1%	93.6% $\pm$ 0.1%	98.8% $\pm$ 0.5%	98.2% $\pm$ 0.5%	96.8% $\pm$ 0.2%	92.1% $\pm$ 0.3%	81.7% $\pm$ 0.6%
FedAvg	95.2% $\pm$ 0.4%	95.2% $\pm$ 0.3%	99.4% $\pm$ 0.1%	98.8% $\pm$ 0.2%	97.7% $\pm$ 0.0%	94.1% $\pm$ 0.7%	85.8% $\pm$ 1.0%
FedProx	94.2% $\pm$ 0.4%	94.2% $\pm$ 0.4%	99.4% $\pm$ 0.2%	98.7% $\pm$ 0.3%	97.6% $\pm$ 0.3%	93.3% $\pm$ 0.2%	82.2% $\pm$ 1.0%
GCFL+	94.6% $\pm$ 0.4%	94.6% $\pm$ 0.3%	99.3% $\pm$ 0.1%	98.7% $\pm$ 0.1%	97.2% $\pm$ 0.4%	93.2% $\pm$ 0.5%	84.4% $\pm$ 1.5%



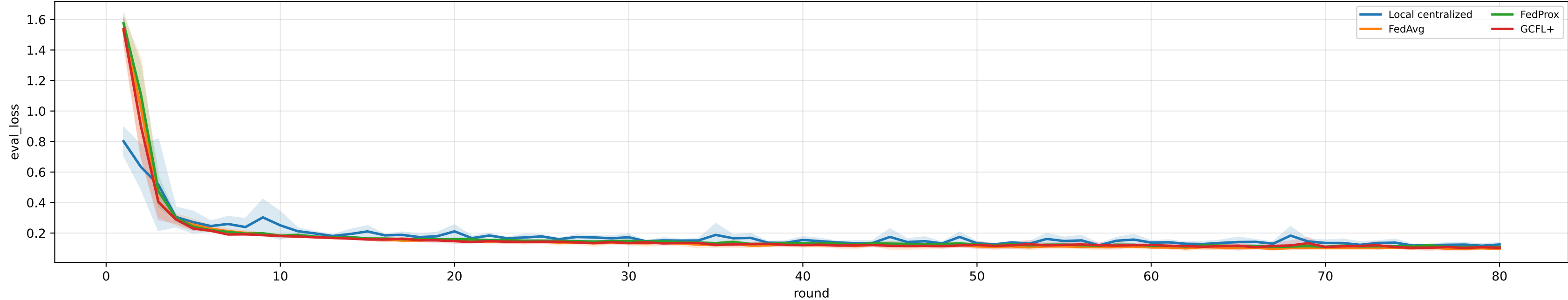
Controlled masking level: High | family aggregated / pooled performance |
family=masked_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle2 | n_clients=1 | cycle2=visible 5.5%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% |
cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1% |
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

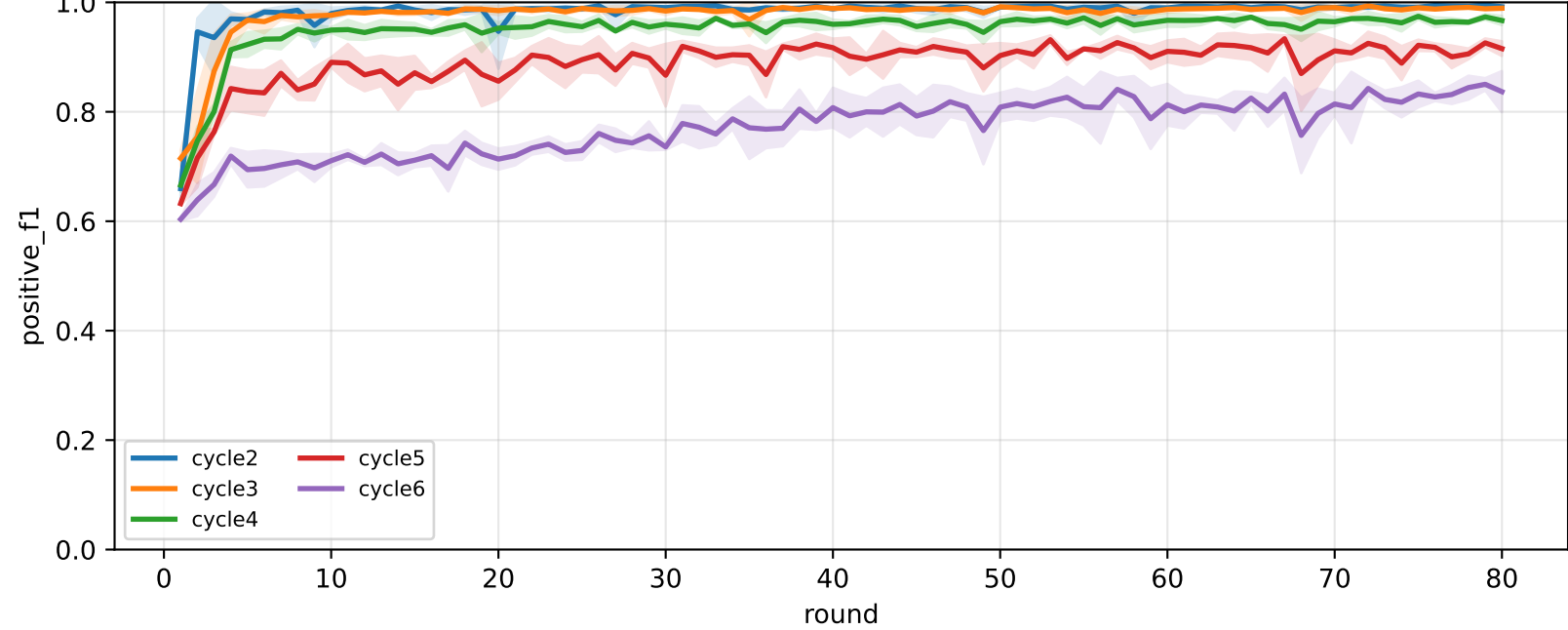
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.1%	94.8% ± 0.1%	99.3% ± 0.0%	98.8% ± 0.2%	97.8% ± 0.2%	93.9% ± 0.4%	84.5% ± 1.0%
FedAvg	95.4% ± 0.3%	95.3% ± 0.3%	99.3% ± 0.1%	98.8% ± 0.3%	97.9% ± 0.1%	94.4% ± 0.8%	86.0% ± 1.0%
FedProx	94.4% ± 0.2%	94.4% ± 0.3%	99.4% ± 0.2%	98.6% ± 0.3%	97.8% ± 0.3%	93.5% ± 0.2%	82.5% ± 0.9%
GCFL+	94.6% ± 0.1%	94.5% ± 0.2%	99.3% ± 0.2%	98.8% ± 0.1%	97.3% ± 0.3%	92.9% ± 1.3%	84.5% ± 0.9%

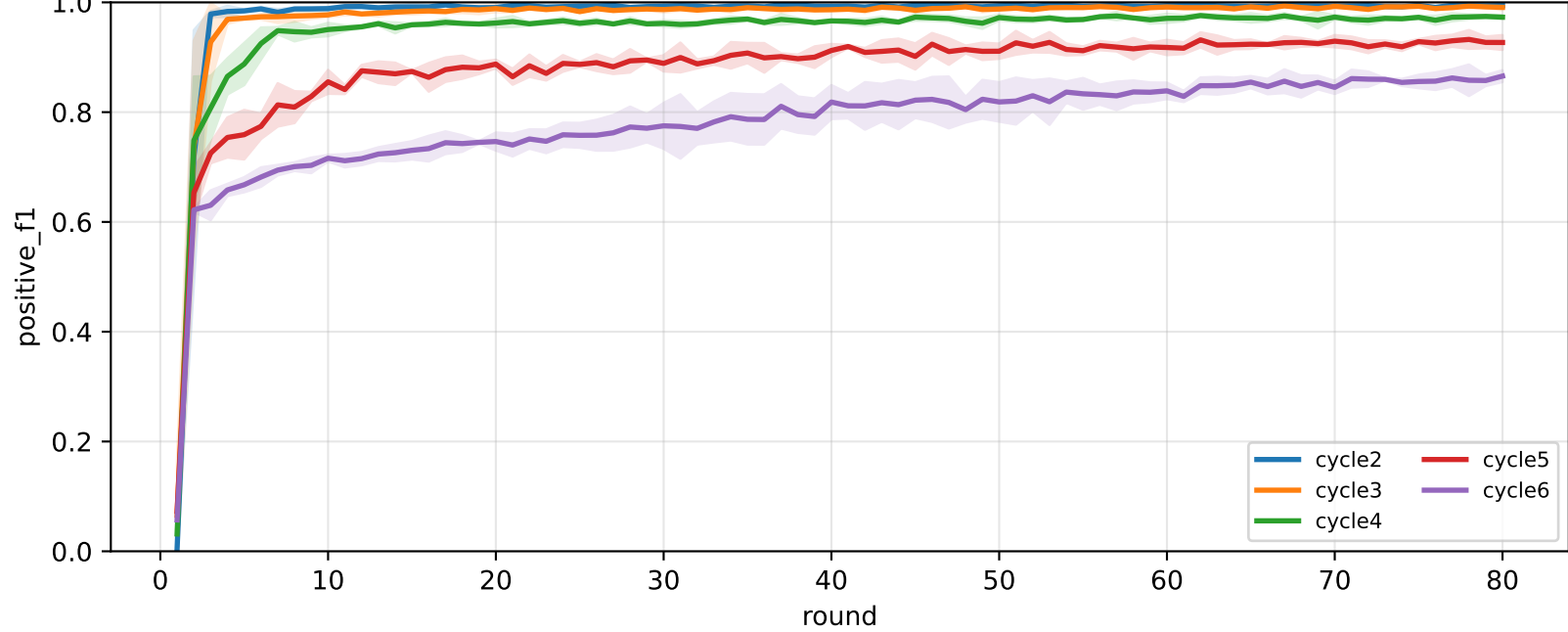
Validation loss comparison



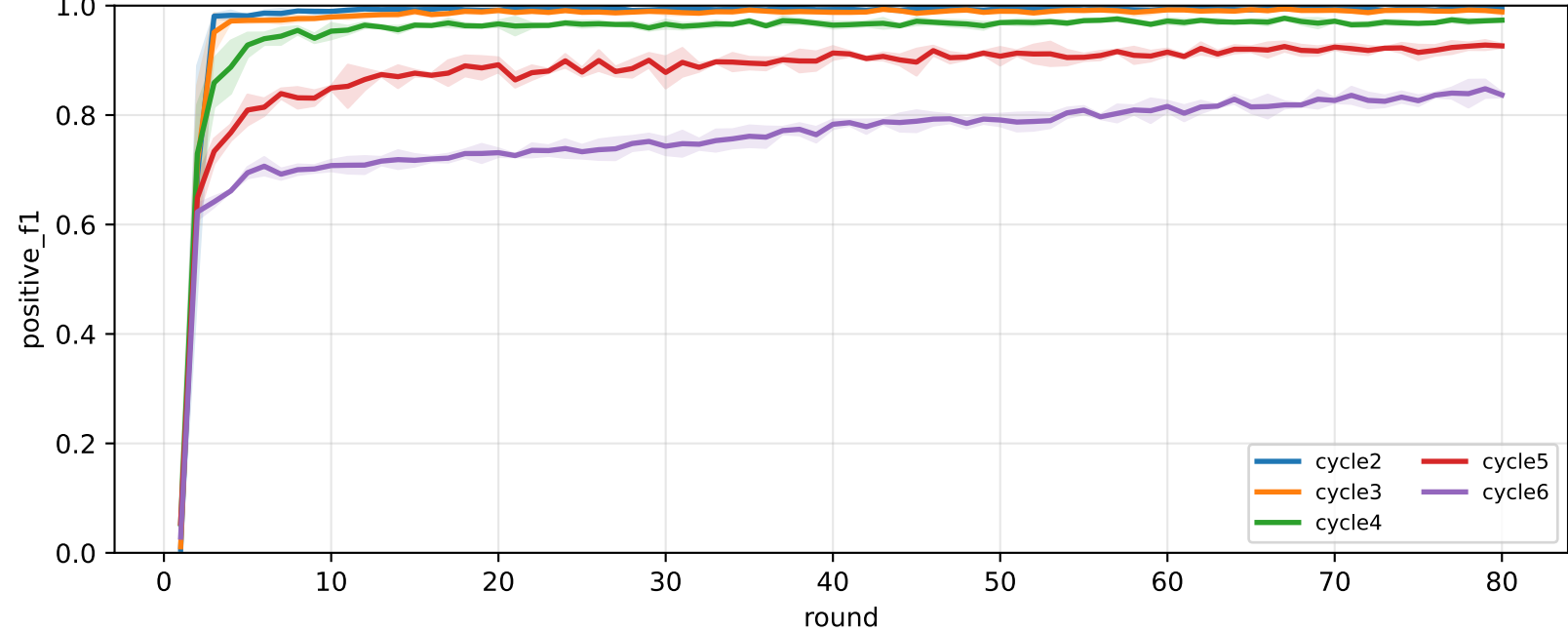
Pooled local centralized positive F1



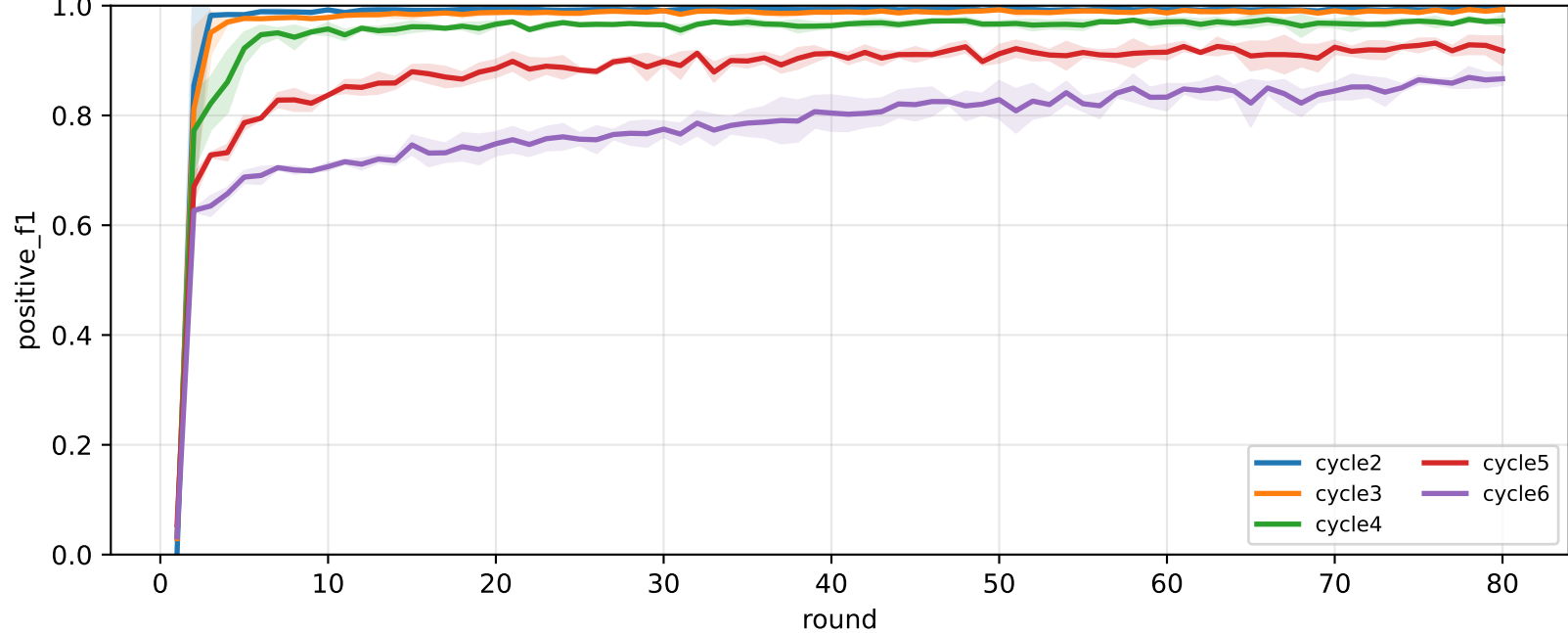
Pooled FedAvg positive F1



Pooled FedProx positive F1



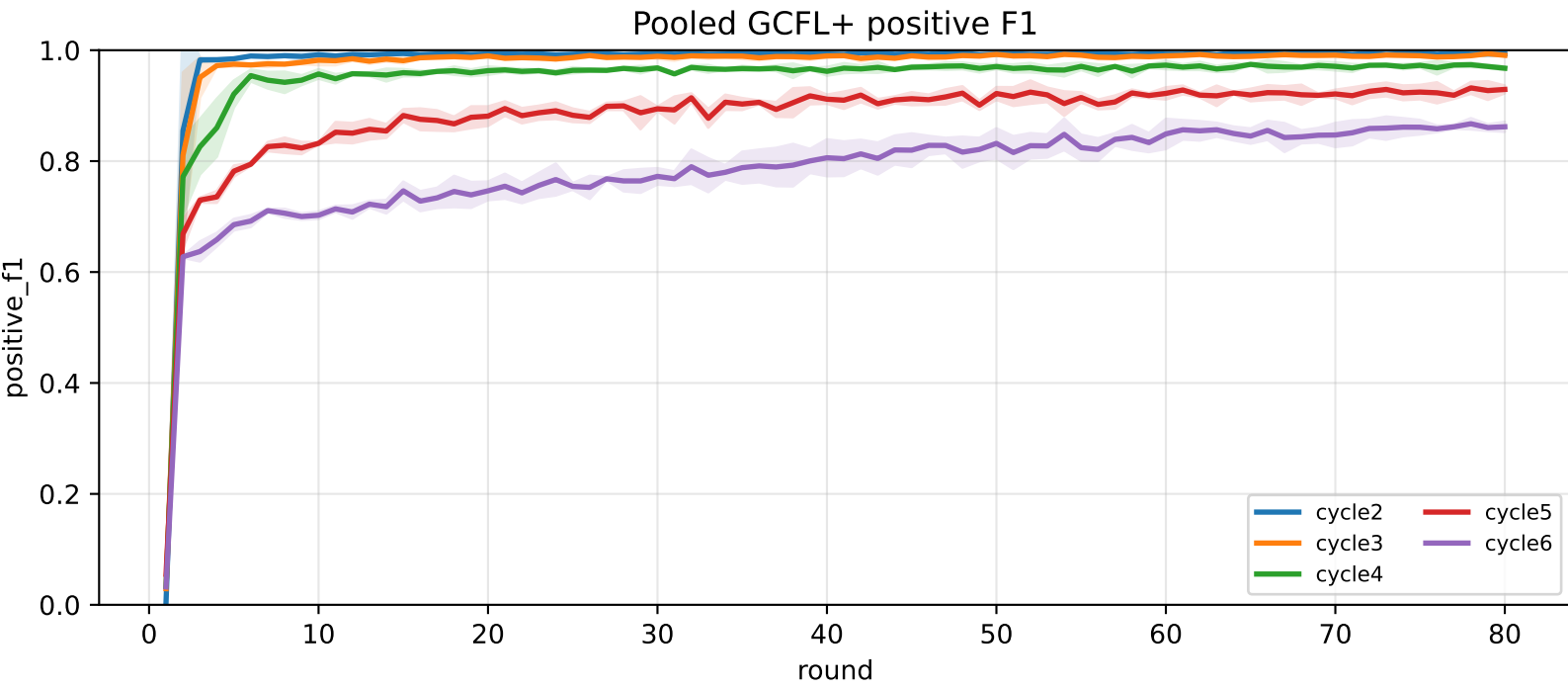
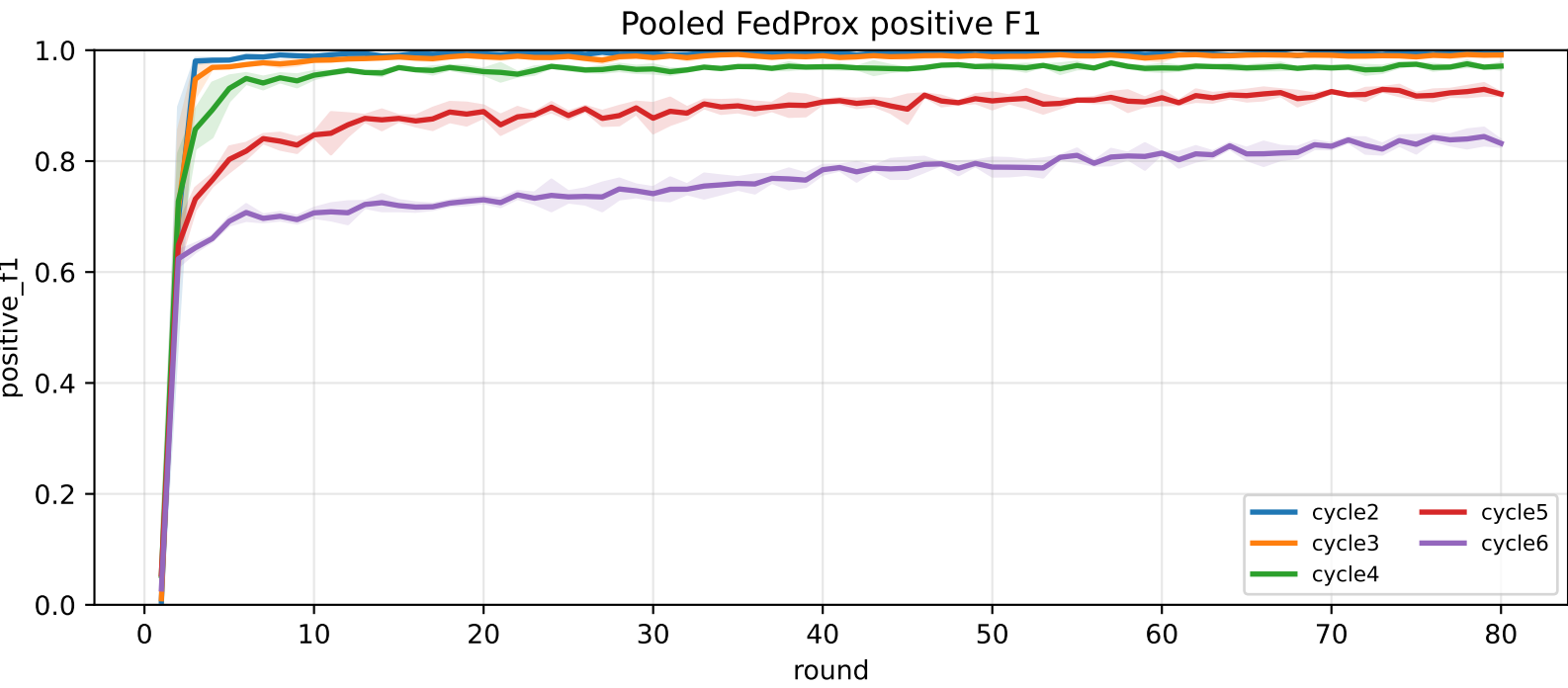
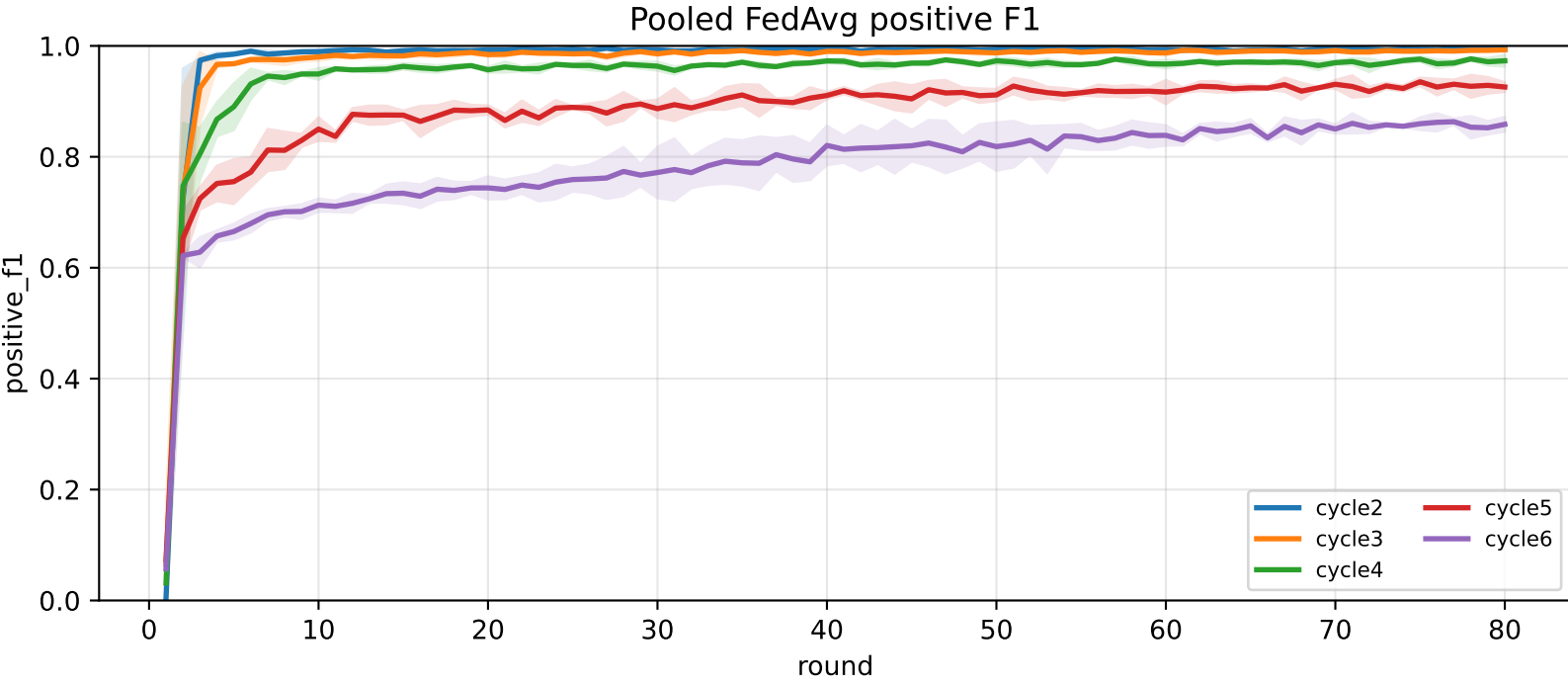
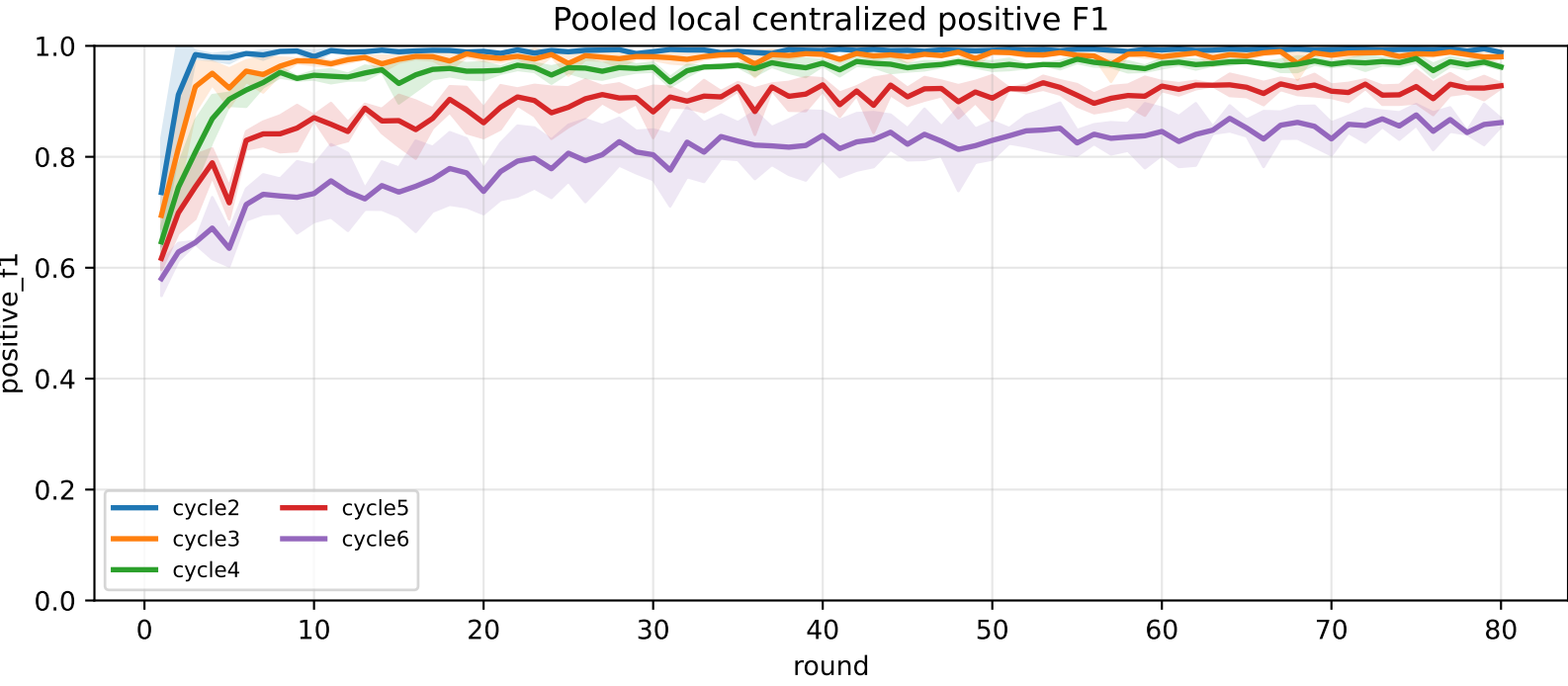
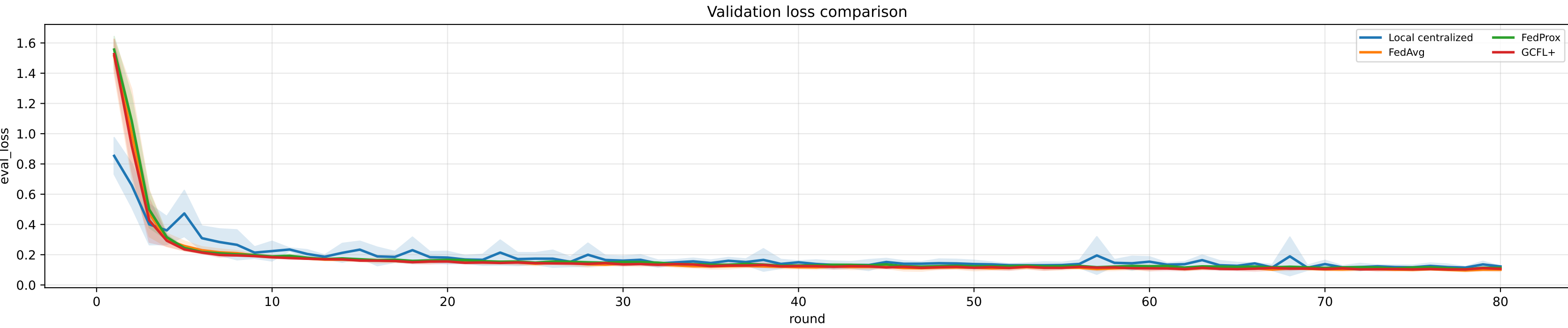
Pooled GCFL+ positive F1



Controlled masking level: High | family aggregated / pooled performance |
family=masked_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle3 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 5.8%±0.0% / true 29.1% |
cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1% |
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.1% ± 0.8%	95.0% ± 0.8%	99.3% ± 0.1%	98.2% ± 0.9%	97.5% ± 0.5%	94.3% ± 0.4%	85.9% ± 2.5%
FedAvg	95.2% ± 0.4%	95.1% ± 0.4%	99.6% ± 0.2%	98.9% ± 0.1%	97.7% ± 0.2%	93.8% ± 0.9%	85.5% ± 0.6%
FedProx	94.2% ± 0.1%	94.2% ± 0.1%	99.5% ± 0.2%	98.8% ± 0.1%	97.6% ± 0.2%	93.1% ± 0.2%	81.9% ± 0.4%
GCFL+	94.8% ± 0.3%	94.7% ± 0.3%	99.2% ± 0.1%	98.6% ± 0.4%	97.4% ± 0.1%	93.9% ± 0.4%	84.6% ± 1.7%



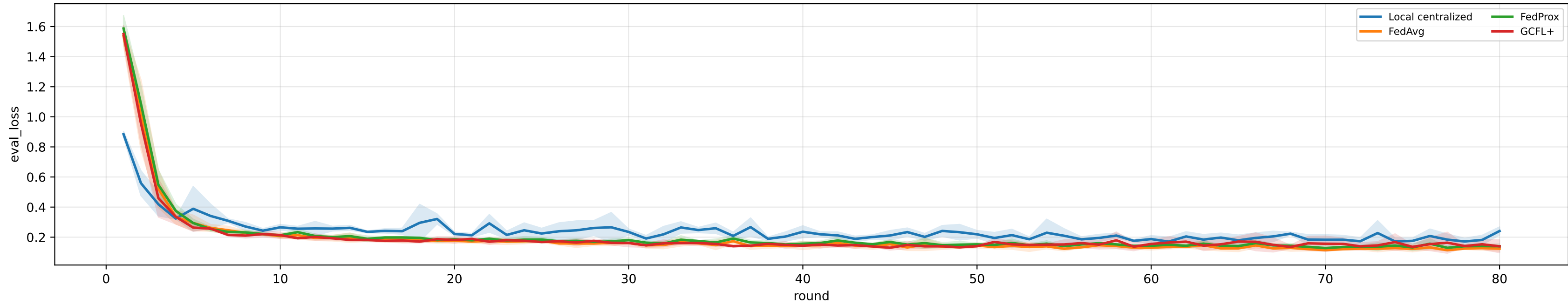
Controlled masking level: High | family aggregated / pooled performance |
family=masked_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle4 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% |
cycle4=visible 5.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1% |
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

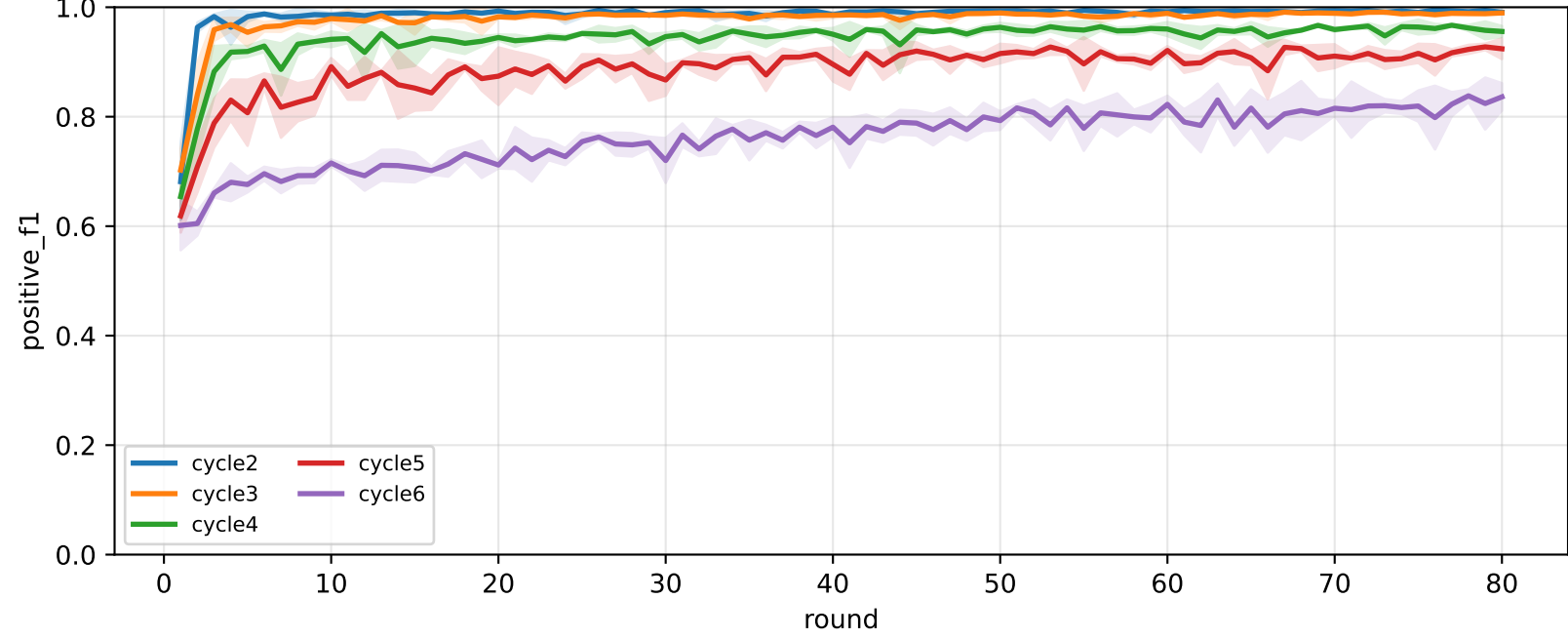
Best logged test performance (mean ± std over seeds)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.4% ± 0.2%	93.5% ± 0.2%	99.2% ± 0.0%	98.2% ± 0.5%	96.5% ± 0.8%	92.3% ± 0.9%	81.1% ± 1.2%
FedAvg	95.3% ± 0.4%	95.3% ± 0.4%	99.2% ± 0.3%	99.0% ± 0.3%	97.7% ± 0.2%	94.2% ± 0.8%	86.3% ± 1.2%
FedProx	94.4% ± 0.5%	94.4% ± 0.5%	99.2% ± 0.3%	98.9% ± 0.4%	97.7% ± 0.4%	93.4% ± 0.5%	82.5% ± 1.3%
GCFL+	95.1% ± 0.2%	95.0% ± 0.1%	99.3% ± 0.1%	98.7% ± 0.3%	97.4% ± 0.9%	93.7% ± 0.4%	86.0% ± 1.6%

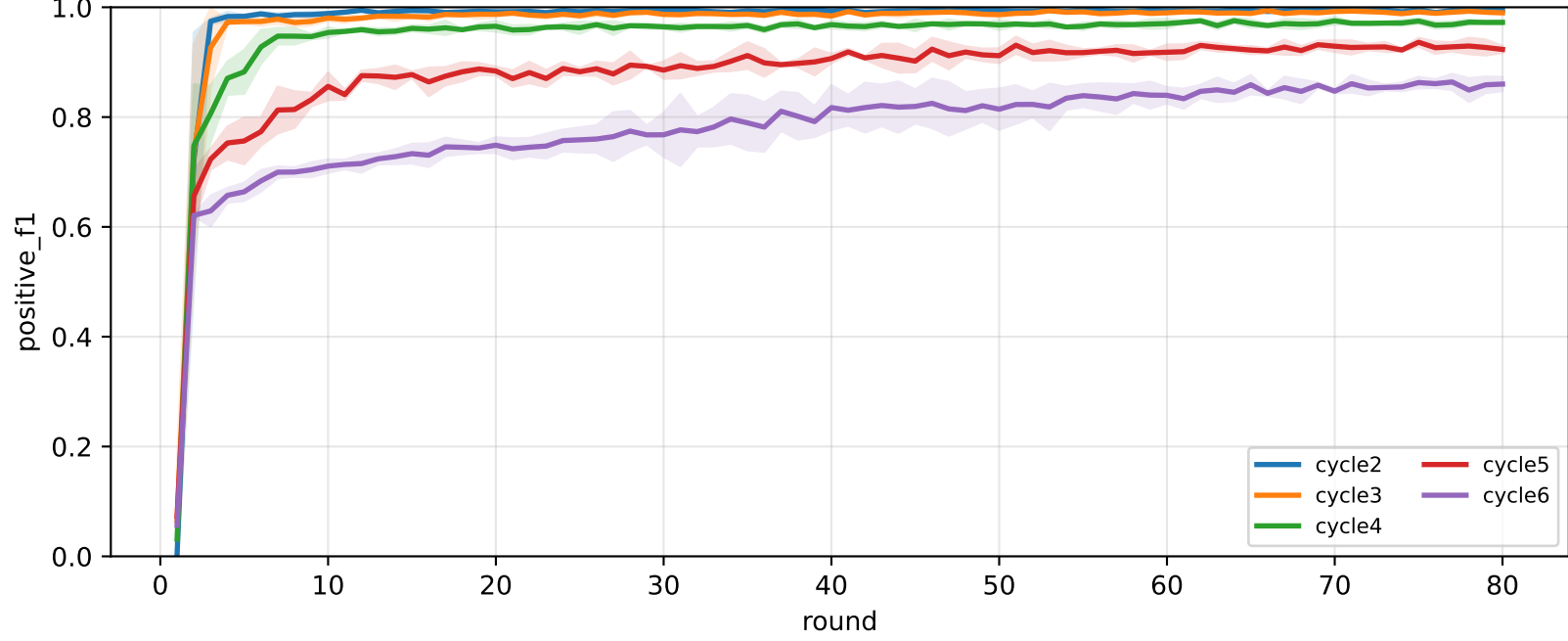
Validation loss comparison



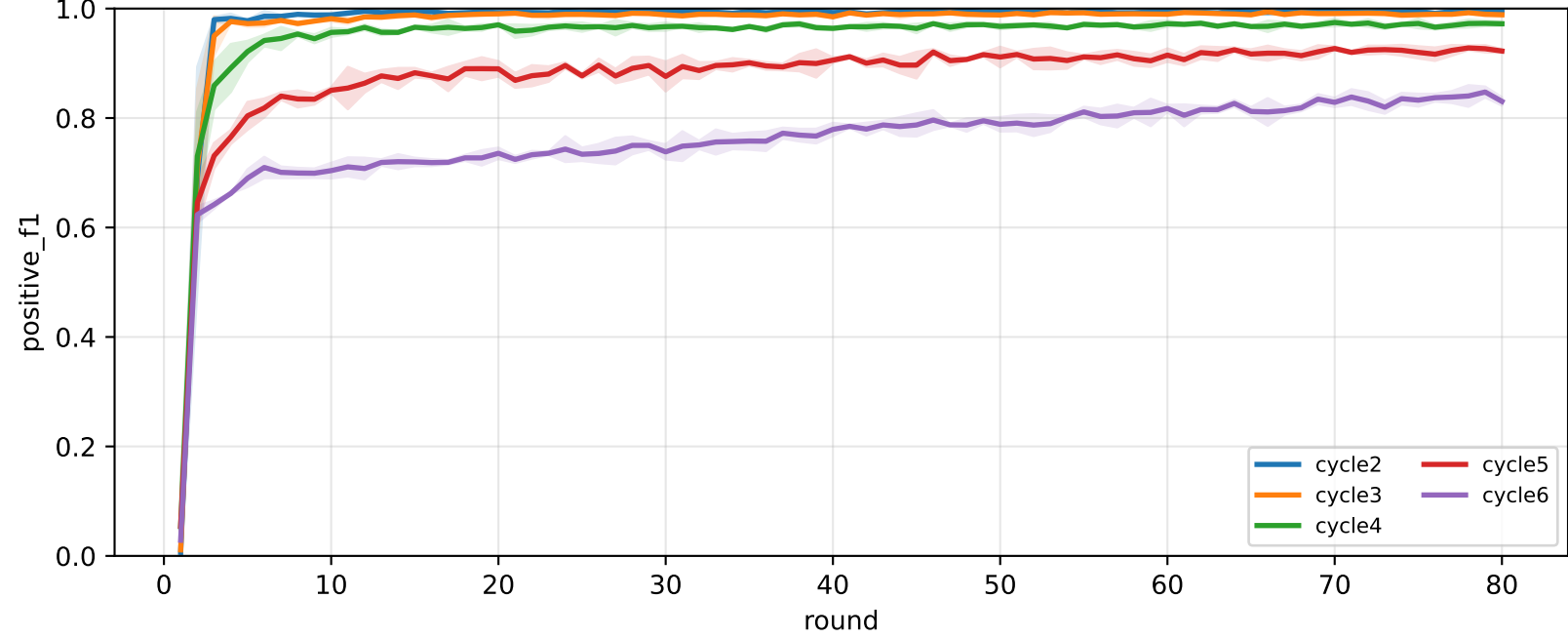
Pooled local centralized positive F1



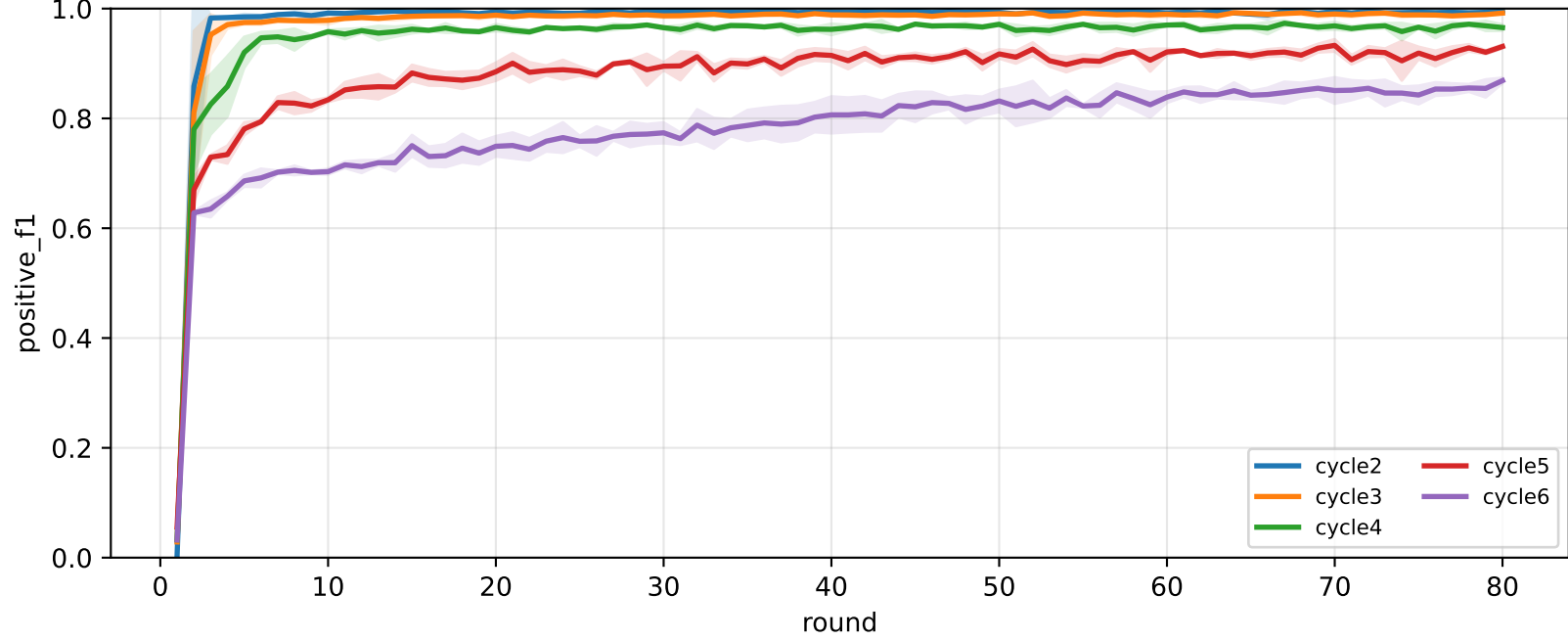
Pooled FedAvg positive F1



Pooled FedProx positive F1



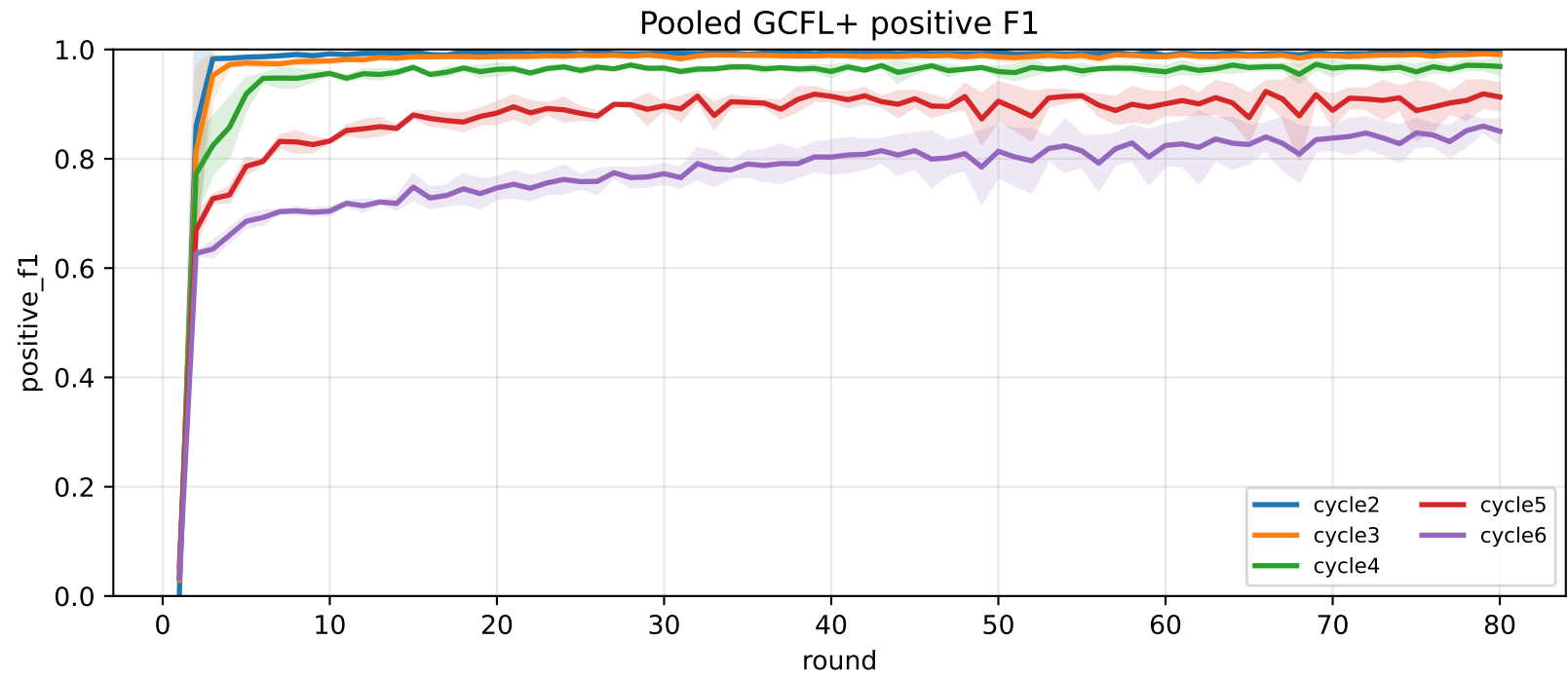
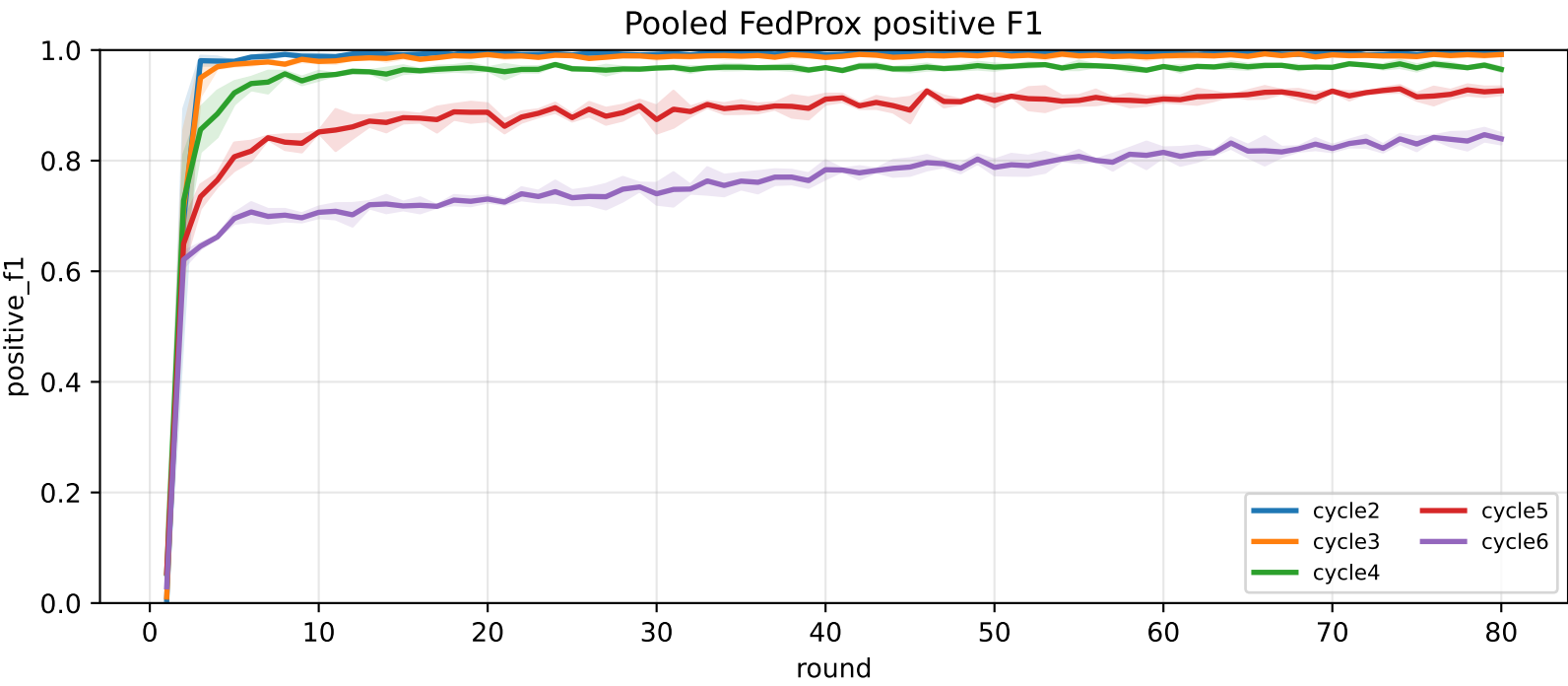
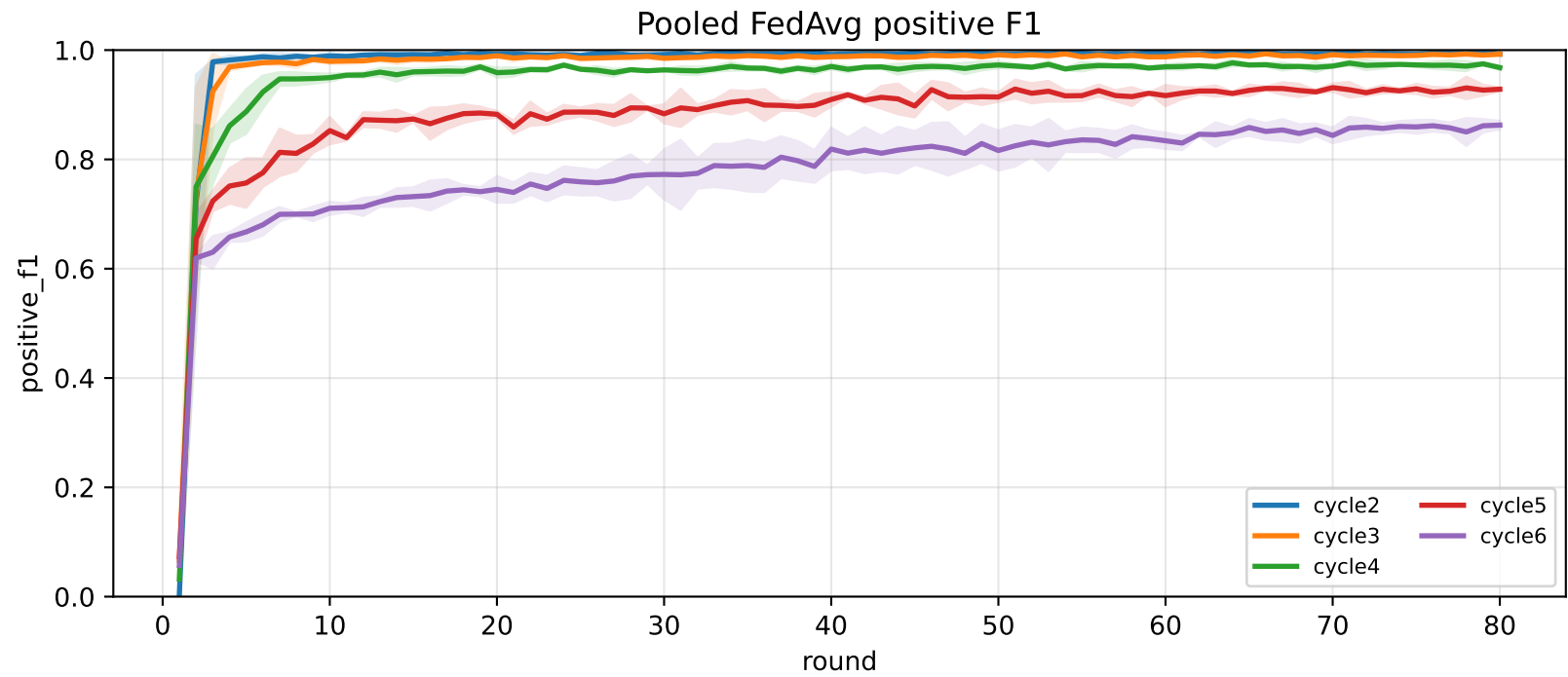
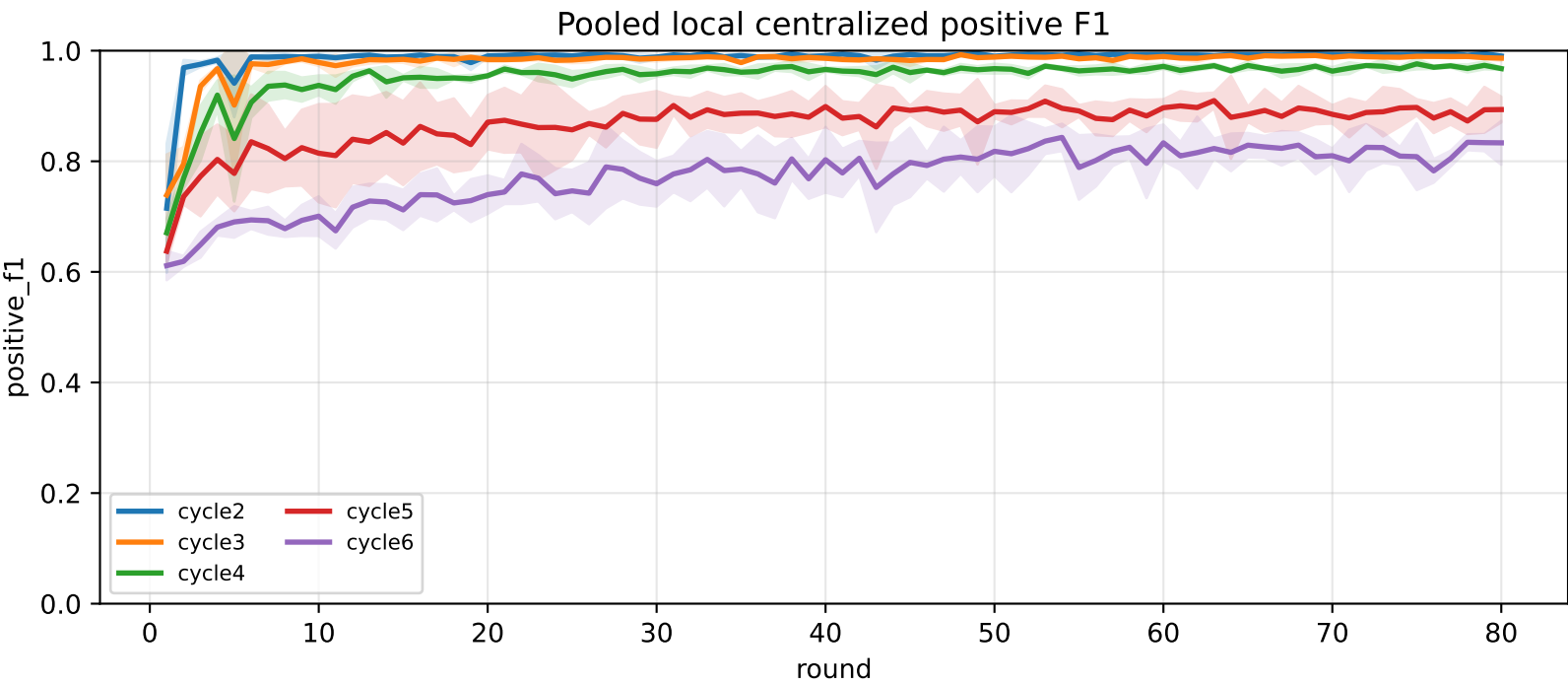
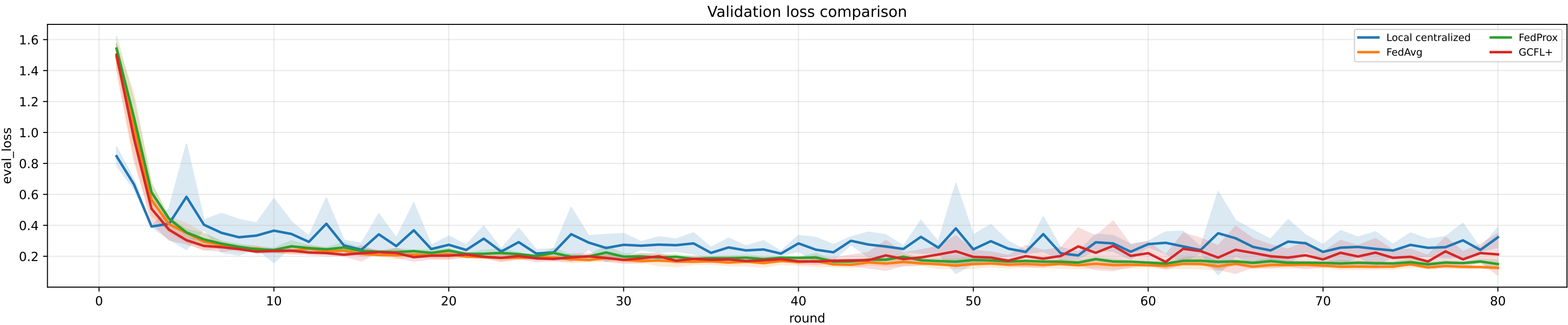
Pooled GCFL+ positive F1



Controlled masking level: High | family aggregated / pooled performance |
family=masked_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle5 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% |
cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 5.2%±0.0% / true 26.2% | cycle6=visible 26.1%±0.0% / true 26.1% |
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.9% ± 0.5%	92.9% ± 0.4%	99.0% ± 0.2%	98.0% ± 0.4%	95.5% ± 0.4%	90.0% ± 1.4%	82.1% ± 1.9%
FedAvg	95.2% ± 0.5%	95.1% ± 0.5%	99.3% ± 0.3%	98.8% ± 0.3%	97.6% ± 0.2%	94.2% ± 0.8%	85.8% ± 1.3%
FedProx	94.2% ± 0.5%	94.2% ± 0.5%	99.2% ± 0.3%	98.6% ± 0.4%	97.7% ± 0.4%	93.4% ± 0.3%	82.1% ± 1.1%
GCFL+	93.6% ± 1.1%	93.6% ± 1.0%	99.3% ± 0.3%	98.6% ± 0.3%	96.6% ± 0.9%	91.6% ± 1.7%	82.2% ± 2.4%



Controlled masking level: High | family aggregated / pooled performance |
family=masked_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=masked_cycle6 | n_clients=1 | cycle2=visible 27.6%±0.0% / true 27.6% | cycle3=visible 29.1%±0.0% / true 29.1% |
cycle4=visible 26.2%±0.0% / true 26.2% | cycle5=visible 26.2%±0.0% / true 26.2% | cycle6=visible 5.2%±0.0% / true 26.1% |
masking=high | mask_fraction=0.80 | task_jsd_mean=0.069 | task_jsd_median=0.069 | task_jsd_max=0.073 |
family_centroid_jsd_mean=0.069

Best logged test performance (mean ± std over seeds)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.5% ± 0.6%	91.7% ± 0.4%	97.3% ± 2.5%	97.9% ± 0.5%	96.8% ± 0.5%	90.4% ± 2.2%	76.1% ± 2.5%
FedAvg	95.1% ± 0.3%	95.0% ± 0.3%	99.5% ± 0.2%	98.6% ± 0.3%	97.5% ± 0.3%	94.0% ± 0.6%	85.5% ± 1.0%
FedProx	94.1% ± 0.6%	94.1% ± 0.6%	99.4% ± 0.3%	98.5% ± 0.4%	97.4% ± 0.7%	93.3% ± 0.3%	82.1% ± 1.3%
GCFL+	94.9% ± 0.4%	94.8% ± 0.4%	99.4% ± 0.2%	98.8% ± 0.3%	97.3% ± 0.5%	94.0% ± 0.5%	84.8% ± 1.7%

